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Formal Logic

UChicago Political Science Math Camp

Ryan Gibbons & Luke Cain

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A Justification of Social Science

- ▶ Social science as an *explanatory pursuit*
- ▶ Evidentiary collection and analysis provides a *means* to positive explanations
- ▶ Unexplained phenomena are unanswered questions

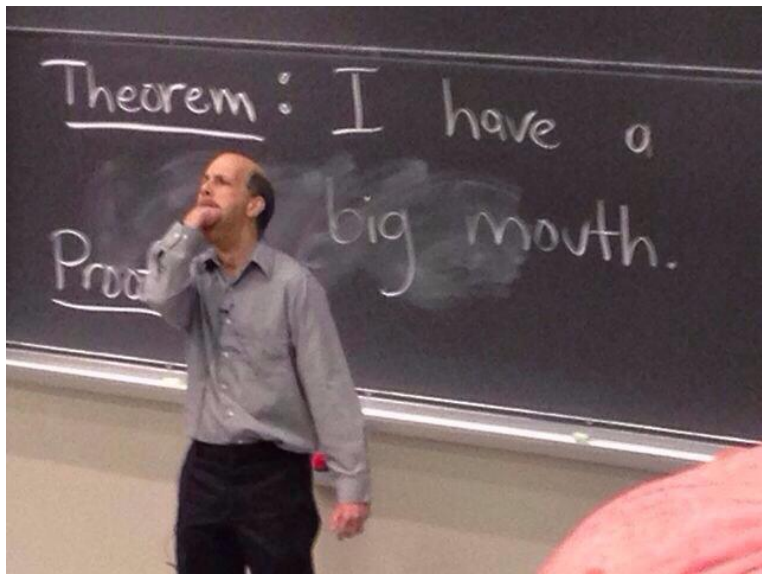
A central motivation: we use empirical information to produce theoretical insights, and we want our theoretical insights to be *maximally applicable* while retaining accuracy

Why Proofs?

It is almost always easier to definitively observe a phenomenon than it is to definitively explain it.

If we cannot definitively explain it, we have not answered the important question (normative argument).

If we cannot definitively explain it, we cannot rule out other potential mechanisms (logical argument).



The Structure of Formal Logic

Formal logic largely focuses on three translations:

1. Empirical observations into informational inputs
2. Informational inputs into theoretical mechanisms
3. Theoretical mechanisms into applied inference and/or prediction

We want to maximize what we can glean from each of these three translations.

Foundational Notation

Formal logic requires formal notation, even beyond our set-building language. A brief overview:

- ▶ $\implies$ “implies”
- ▶ $\iff$ “if and only if” (iff)
- ▶ $\neg$ “not”
- ▶ $\not\implies$ “does not imply.” ($\backslash$ generally means “not.”)

You might occasionally see $\wedge$ for “and,” $\vee$ for “or.” Just write the damn words

An Aside For This Lecture

In this lecture, I will refer to “A,B,C...” as *individual claims*

I will refer to *logical conditions* using greek letters – α , β , etc.

We don't apply any substantive meaning (yet).

Implies

Our strongest operator is $\implies$, typed as `\implies`:

If $A \implies B$, then claim A 's truth lends claim B 's truth.

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For example:

- ▶ $[\text{state}] \text{ follows the "Nordic Model"} \implies [\text{state}] \text{ employs a social democratic welfare system}$
- ▶ $[\text{state}] \text{ is a liberal democracy} \implies [\text{state}] \text{ has a legislature OR } [\text{state}] \text{ is a direct democracy}$
- ▶ $\text{United Nations recognizing } [\text{state}] \implies [\text{state}] \text{ is a country}$

Limits of Implication

Implication is a **unidirectional** relationship: $A \implies B$ does not tell us that $B \implies A$.

Consider the claim $A \implies B$. Then, we have:

- ▶ If A , then B ;
- ▶ If $\neg B$, then $\neg A$

But we do **NOT necessarily** have:

- ▶ If B , then A ;
- ▶ If $\neg A$, then $\neg B$

What do these conditions look like formalized?

Handling Negation (pt. 1)

We use two different “negation” operators, with the following loose applications...

- ▶ We use $/$ over an operation to show that the relationship is **not necessarily** true
- ▶ We use $\neg$ to show that some relationship is **necessarily not** true

For example, the following two condition are **not** equivalent:

1. $A \not\Rightarrow B$

2. $A \Rightarrow \neg B$

Iff

Our strongest operator is $\iff$, or “if and only if.” This implies a **bidirectional relationship**.

If $A \iff B$, then:

1. $A \implies B$, AND
2. $B \implies A$

Neither A nor B can be true on their own.

If we know $A \iff B$ to be true, what do we need to observe to determine either A or B ?

Iff Examples

Iff statements are especially present when defining terms and concepts.

Why?

- ▶ [state] has free & fair elections with uncertain outcomes $\iff$ [state] is a democracy
- ▶ A central bank is entirely independent $\iff$ elected officials cannot control or influence the interest rate
- ▶ A legislative bloc has an outright majority $\iff$ it holds more than half of legislative seats

Logical Grammar

Logical statements often require complexity. We use parentheses to identify an order of operations:

- ▶ $(A \text{ and } B) \implies C$: If A and B , then C .
- ▶ $A \implies (B \text{ and } C)$: If A , then B and C .

This looks simple so far...

Handling Negation (pt. 2)

Logical ordering becomes very important when considering negative conditions or interactions. For example, the following statements are **not** equivalent:

1. $\neg(A \text{ and } B) \implies C$

2. $\neg A \text{ and } \neg B \implies C$

Which of these statements would be more challenging to prove?

Strength of Conditions

Some conditions are harder to meet than others. A helpful consideration is identifying **nesting** conditions: is one condition implies by another?

We can consider a condition α as **stronger** than a condition β if (not iff) $\alpha \implies \beta$.

Strength of Conditions

Some conditions are harder to meet than others. A helpful consideration is identifying **nesting** conditions: is one condition implies by another?

We can consider a condition α as **stronger** than a condition β if (not iff) $\alpha \implies \beta$.

For example, the following conditions have increasing strength:

1. A legislative bloc has plurality control
2. A legislative bloc is the dominant group of a majority coalition
3. A legislative bloc has an outright majority
4. A legislative bloc has a supermajority (can pass any item without failure)

An Interesting Application: Bureaucratic Discretion

A common question in domestic politics regards how much *discretion* principals lend to their agents (see Huber & Shipman 2002).

Central tradeoff: providing discretion allows for both effectiveness and shirking.

But, to evaluate this theory, we must measure bureaucratic discretion...

An Interesting Application: Bureaucratic Discretion

Much of the literature on bureaucratic discretion uses legislative word count as an independent variable (again, see Huber & Shipman 2002). This is a proxy in accordance with the following logic:

- ▶ A bureaucrat facing conditions A and B is more constrained than a bureaucrat facing only A or only B
- ▶ In order to implement conditions A and B , legislators often need to write *at least A plus* some additions to achieve B
- ▶ Therefore, in expectation, longer written laws should imply additional conditions on bureaucratic behavior (and therefore less discretion)

Back to It

Clearly, $\iff$ is a stronger condition than $\implies$. (Why?)

Back to It

Clearly, $\Longleftrightarrow$ is a stronger condition than $\Longrightarrow$. (Why?)

Is $\nleftrightarrow$ a stronger condition than $\Longleftrightarrow$?

Back to It

Clearly, $\iff$ is a stronger condition than $\implies$. (Why?)

Is $\nRightarrow$ a stronger condition than $\iff$?

Is $\nRightarrow$ a stronger condition than $\nRightarrow$?

Back to It

Clearly, $\iff$ is a stronger condition than $\implies$. (Why?)

Is $\nRightarrow$ a stronger condition than $\iff$?

Is $\nRightarrow$ a stronger condition than $\implies$?

Is $\neg \iff \neg$ a stronger condition than $\nRightarrow$?

Nesting: Necessary & Sufficient

We can characterize conditions as **necessary** and/or **sufficient**:

- ▶ Condition α is sufficient for condition β if $\alpha \implies \beta$.
 - ▶ If this is true, what does observing β tell us?
- ▶ Condition α is necessary for condition β if $\neg\alpha \implies \neg\beta$
 - ▶ If this is true, what does observing β tell us?

Which type of condition is stronger?

What Is To Be Done?

To produce meaningful conclusions, we will focus on four fundamental “proofs” using these logical operations:

1. Direct
2. Contrapositive
3. Contradiction
4. Cases

These are general strategies; there's not always a “correct” fit.

Direct Proofs

Direct proofs simply show that multiple conditions are equivalent.

Suppose we want to prove $A \iff B$. Then, a direct proof would need to show both of the following:

- ▶ If A , then B ($A \implies B$)
- ▶ If B , then A ($B \implies A$)

Why Not Direct?

Direct proofs can be great *if we have sufficient information*; we don't always.

For example, suppose we want to prove that every perfect direct democracy¹ has an imbalanced tiebreak institution. Can we prove this directly?

¹Every citizen has equal influence over outcomes

The Contrapositive

Contrapositives invert $\implies$ and allows us to prove via **falsification**.
Namely, $A \implies B$ is equivalent to $\neg B \implies \neg A$.

Suppose we want to prove $A \iff B$, but could not conclude directly that $B \implies A$. Then, a contrapositive proof would need to show:

- ▶ If A , then B (or $A \implies B$)
- ▶ If $\neg B$, then $\neg A$ (or $\neg B \implies \neg A$)

This tends to be an easier proof of $\iff$ statements than the direct route above.

A Contrapositive Example

Can we use a contrapositive proof to prove our above claim (that every perfect direct democracy has an imbalanced tiebreak institution)?

A Contrapositive Example

Can we use a contrapositive proof to prove our above claim (that every perfect direct democracy has an imbalanced tiebreak institution)?

Yes! We just need to show the following: If a state does NOT have an imbalanced tiebreak institution, then it is NOT a perfect direct democracy. We can prove this directly.

Proof by Contradiction

Contradiction is a powerful proof technique to demonstrate that a claim is true *in all cases*.

We start with a claim that contradicts what we want to prove. If we want to prove $A \iff B$, we then start with: “Suppose not. Then...”

Think of contradiction as a rabbit-hole proof: expand conditions as far as we can, until something breaks

Contradiction in Abstract

Let's expand upon the arbitrary proof by contradiction (proving that $A \iff B$). The proof might proceed:

1. Suppose not.
2. Then, either $A \not\Rightarrow B$ or $B \not\Rightarrow A$.
3. Then, either there must exist a case satisfying only A and not B or only B and not A .
4. *Disprove either claim.*

Using Contradiction: MVT

The Median Voter Theorem suggests that candidate platforms will converge to the preference of the median voter:

Suppose there are two candidates A and B who establish policy positions along a single dimension (canonically, a preferred tax rate). Then, if A and B are only motivated by winning office, they will both establish policy positions equivalent to the preference of the mathematically median voter.

Why All The Excitement?

These proof structures have gotten increasingly more complex – but, they have widened the range of relationships we are examining, and we are usually just looking for one to “break.”

From $A \iff B$, we've extended to include:

- ▶ Direct: $A \implies B, B \implies A$
- ▶ Contrapositive: $A \implies B, \neg B \implies \neg A$
- ▶ Contradiction: $A \not\implies B, B \not\implies A$

Proof structures allow us to cast wider nets when fishing for claims.

Proof by Cases

Not necessarily mutually exclusive from the above, but an important consideration: oftentimes, we may only have a finite number of possible cases, and can use that to our advantage.

Suppose we want to prove that every party represented in Congress holds a primary. We have two options:

1. Prove that in order to be represented in Congress, a party must have held a primary
2. Just look at the parties represented in Congress (2ish), and see whether or not they held a primary

Proof By Cases

One common case structure we can often use as a shortcut is derived from ordering. Suppose we have two parties, LUKE and RYAN. Then, one of the following **MUST** be true w.r.t. vote counts:

1. $\#LUKE > \#RYAN$
2. $\#LUKE < \#RYAN$ (most likely)
3. $\#LUKE = \#RYAN$

Not every case needs to be proven the same way...

Axiomatic Truths

Our goal is usually to prove “as far down as possible” – but, we need some initial foundations.

Axioms are things we hold true without proofs; they are generally extremely basic.